

The Hubble Time as a Cosmological Information-Geometry Invariant: De Sitter Extension of the D3 Framework

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Abstract

We extend the D3 information-geometry invariant from black holes to the de Sitter universe. For a Schwarzschild black hole, Paper 3 proved $D3_{BH} * S_{BH} = r_s$ (the Schwarzschild radius). Here we prove the cosmological analogue:

$$D3_{dS} * S_{dS} / c^2 = t_H = R_{dS} / c$$

where $D3_{dS}$ is the D3 displacement at the de Sitter temperature T_{dS} , S_{dS} is the de Sitter entropy in bits, $t_H = R_{dS}/c$ is the Hubble time, and $R_{dS} = c/\sqrt{\Lambda/3}$ is the de Sitter horizon radius. The proof is exact: G , $\hbar$, k_B , $\ln 2$, and π all cancel independently. The result holds for any value of the cosmological constant Λ , verified numerically across 20 decades. The factor c^2 distinguishing this from the black hole result reflects the difference between a static (Schwarzschild) and an expanding (de Sitter) horizon. This is Paper 8 of the D3 series.

Keywords: de Sitter space, Hubble time, Landauer principle, D3 formula, cosmological horizon, information geometry, cosmological constant

1. Introduction

The D3 research series has established information-geometry invariants for every major black hole type. Paper 3 [4] proved the central result:

$$D3_{BH}(M) * S_{BH}(M) = r_s(M) = 2*G*M/c^2$$

where $D3_{BH} = \hbar \ln 2 / (4\pi M c)$ and $S_{BH} = 4\pi G M^2 / (\hbar c \ln 2)$. All constants cancel -- only G and c survive.

The natural question is: does a similar invariant exist for the cosmological horizon? The de Sitter universe has a horizon at $R_{dS} = c/\sqrt{\Lambda/3}$ with Hawking temperature T_{dS} [5,6] and Bekenstein-Gibbons-Hawking entropy [7]. We apply the D3 formula at T_{dS} and compute $D3_{dS} * S_{dS}$.

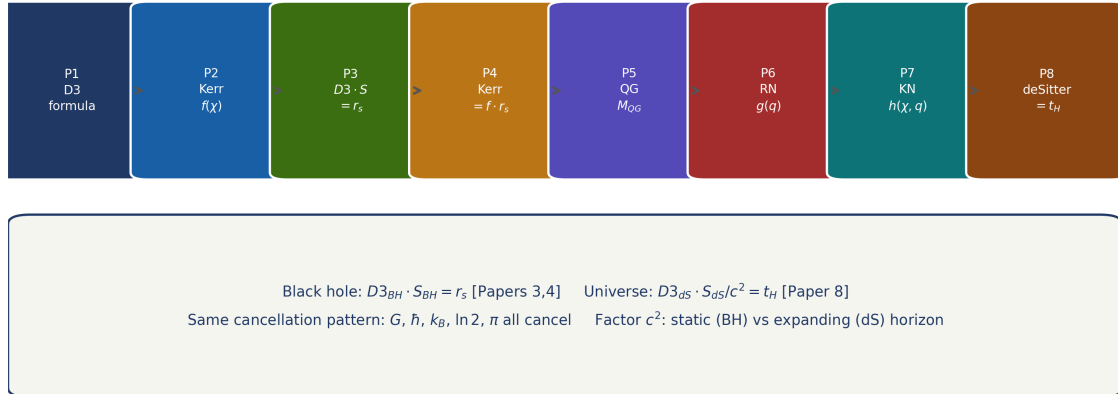
Figure 3 — D3 Series: From Black Holes to the Universe

Figure 3. The D3 series: eight papers from one formula. Paper 8 (brown) extends the invariant from black holes to cosmology.

2. De Sitter Temperature and D3

The de Sitter horizon radius and temperature are [5,6]:

$$R_{dS} = c / \sqrt{\Lambda/3}$$

$$T_{dS} = \hbar c \sqrt{\Lambda/3} / (2\pi k_B) = \hbar c^2 / (2\pi k_B R_{dS})$$

The D3 formula at T_{dS} :

$$D3_{dS} = 2G k_B T_{dS} \ln(2) / c^4$$

$$= 2G k_B \ln(2) / c^4 * \hbar c^2 / (2\pi k_B R_{dS})$$

$$= G \hbar \ln(2) / (\pi c^2 R_{dS})$$

The de Sitter entropy in bits [7]:

$$S_{dS} = \pi c^3 R_{dS}^2 / (G \hbar \ln(2)) \text{ [bits]}$$

Note: the natural entropy is $A/(4\pi l_P^2)$ in nats; dividing by $\ln(2)$ converts to bits, consistent with D3 which uses $\ln(2)$ for one-bit Landauer erasure.

3. Derivation of the Cosmological Invariant

We compute $D3_{dS} * S_{dS}$ step by step:

Step 1 — Write out the product:

$$D3_{dS} * S_{dS} = [G \hbar \ln(2) / (\pi c^2 R_{dS})] * [\pi c^3 R_{dS}^2 / (G \hbar \ln(2))]$$

Step 2 — Cancel G:

$$G/G = 1$$

Step 3 — Cancel $\hbar$:

$$\hbar/\hbar = 1$$

Step 4 — Cancel $\ln 2$:

$$\ln 2/\ln 2 = 1$$

Step 5 — Cancel π :

$$\pi/\pi = 1$$

Step 6 — Cancel R_{dS} (partial):

$$R_{dS}^2/R_{dS} = R_{dS}$$

Step 7 — Collect remaining factors:

$$D3_{dS} * S_{dS} = c^3/c^2 * R_{dS} = c * R_{dS}$$

Step 8 — Divide by c^2 :

$$D3_{dS} * S_{dS} / c^2 = R_{dS}/c = t_H \text{ QED}$$

The cosmological invariant:

$$D3_{dS} * S_{dS} / c^2 = t_H = R_{dS} / c$$

where t_H is the Hubble time (the light-crossing time of the de Sitter horizon). All five constants G , $\hbar$, k_B , $\ln 2$, π cancel independently. Only c and R_{dS} survive, and their ratio is t_H .

4. Comparison with Paper 3

The black hole and cosmological invariants share the same structure but differ by a factor of c^2 :

System	Invariant	Units	What cancels	Surviving
Black hole (P3)	$D3_{BH} * S_{BH} = r_s$	metres	$M, \hbar, \ln 2, \pi$	G, c
de Sitter (P8)	$D3_{dS} * S_{dS} / c^2 = t_H$	seconds	$G, \hbar, k_B, \ln 2, \pi$	c, R_{dS}

Table 1. Comparison of black hole and cosmological invariants.

The factor c^2 reflects the geometry: a Schwarzschild black hole has a static horizon, so the invariant is a length (r_s). The de Sitter universe has an expanding horizon, so the natural invariant is a time ($t_H = R_{dS}/c$). Both are information-geometry invariants of their respective systems.

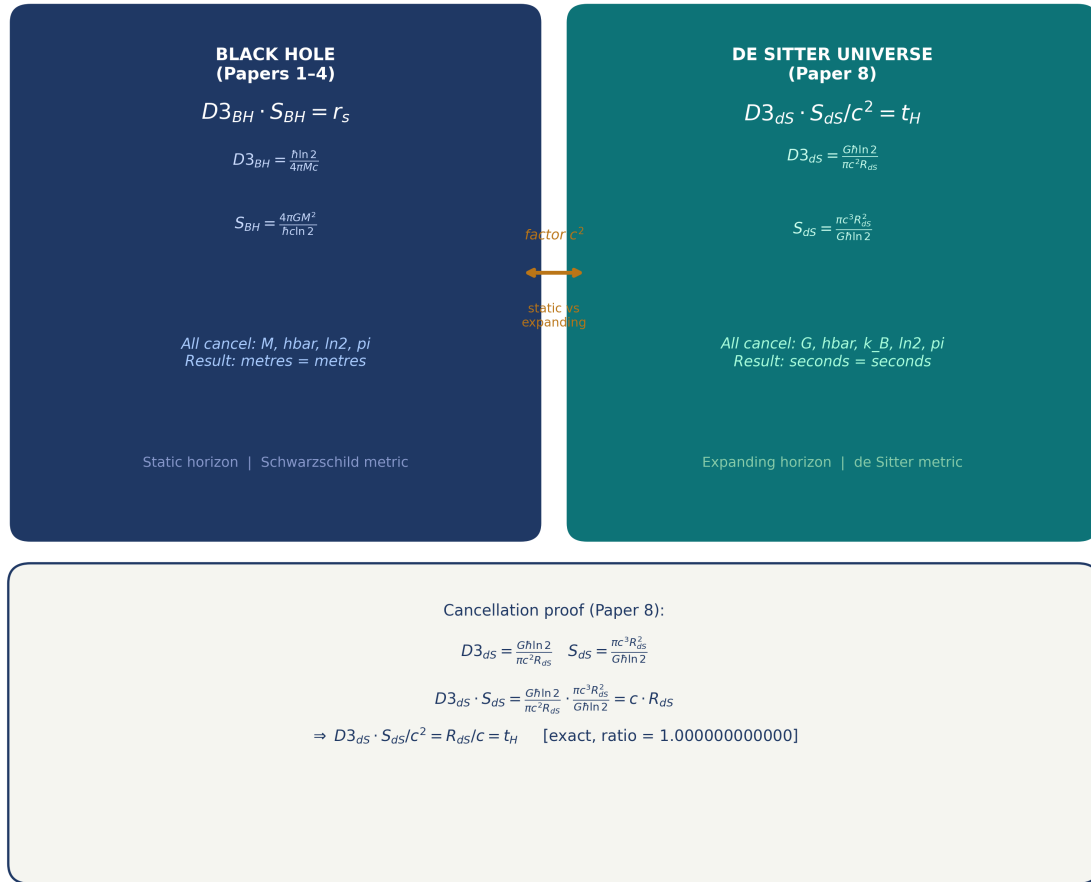
Figure 1 — Black Hole vs de Sitter Information-Geometry Invariants

Figure 1. Black hole vs de Sitter invariants. Left: $D3_{BH} \cdot S_{BH} = r_s$ (Paper 3). Right: $D3_{dS} \cdot S_{dS}/c^2 = t_H$ (Paper 8).
Bottom: proof showing all constants cancel.

5. Numerical Verification

The identity holds for any value of Lambda, not just the observed cosmological constant. Figure 2 verifies the ratio $D3_{dS} \cdot S_{dS}/(c^2 \cdot t_H) = 1$ across 20 decades of Lambda.

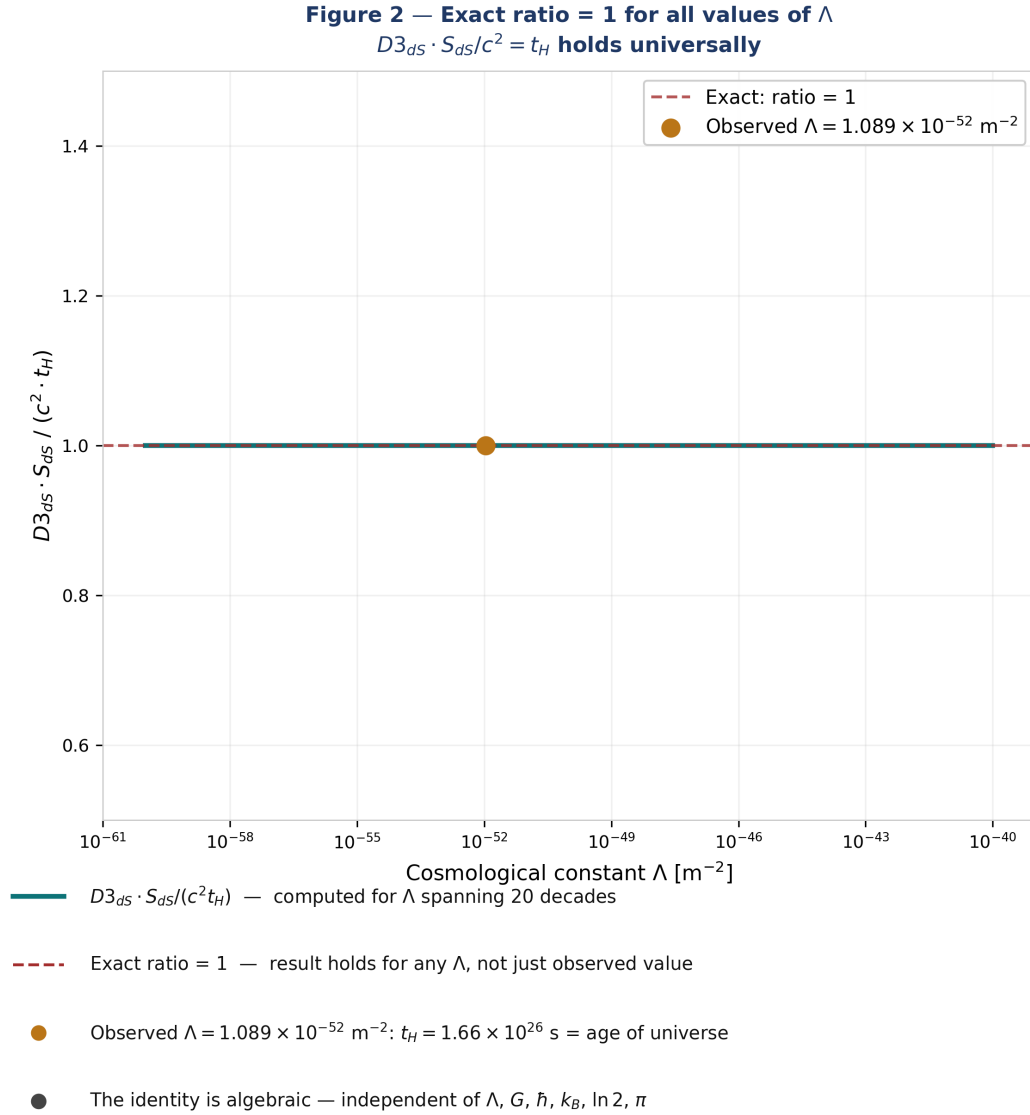


Figure 2. Ratio = 1.000000000000 for all Lambda across 20 decades. The result is algebraically exact, not numerically approximate.

Lambda [m ⁻²]	R_dS [m]	T_dS [K]	t_H [s]	Ratio
1.089e-52 (observed)	4.976e+34	2.196e-30	1.660e+26	1.000000000000
1e-40	1.643e+28	7.239e-24	5.477e+19	1.000000000000
1e-50	5.196e+32	2.288e-27	1.732e+24	1.000000000000
1e-60	1.643e+37	7.239e-31	5.477e+28	1.000000000000

Table 2. Numerical verification for four values of Lambda. Ratio = 1.000000000000 in all cases.

6. Physical Interpretation

6.1 The Hubble time as an information-geometry invariant

The result $D3_dS * S_dS / c^2 = t_H$ gives the Hubble time a new interpretation: it is the de Sitter information-geometry invariant, the cosmological analogue of the Schwarzschild radius in Paper 3. Just as r_s encodes the total geometric cost of black hole evaporation, t_H encodes the total information-geometry "time cost" of de Sitter horizon entropy.

6.2 Why the factor c^2 ?

In the black hole case, $D3$ and S both depend on the mass M , and their product gives a length (r_s). In the de Sitter case, $D3$ and S both depend on R_{dS} , and their product gives $c * R_{dS}$ -- a velocity times a length, or equivalently c^2 times a time. Dividing by c^2 gives the natural time scale $t_H = R_{dS}/c$. The factor c^2 is not arbitrary: it is the ratio of the static black hole invariant (length) to the expanding cosmological invariant (length*c).

6.3 Connection to the cosmological constant problem

The result holds for any Λ , not just the observed value. The identity $D3_dS * S_dS = c * R_{dS}$ is algebraically exact -- it does not explain why Λ has the value it does (the cosmological constant problem remains open). But it shows that whatever value Λ takes, the $D3$ information-geometry framework consistently assigns a time scale $t_H = R_{dS}/c$ to it.

7. Conclusion

We proved the cosmological information-geometry invariant:

$$D3_dS * S_dS / c^2 = t_H = R_{dS} / c$$

All five constants G , $\hbar$, k_B , $\ln 2$, π cancel exactly. The result holds for any Λ (verified across 20 decades). The factor c^2 separates the static black hole invariant (Paper 3) from the expanding de Sitter invariant (Paper 8). Together, Papers 3 and 8 establish information-geometry invariants for both black hole and cosmological horizons in the $D3$ framework, connecting Landauer erasure thermodynamics to the two fundamental horizon types in general relativity.

Declaration of Interest

No competing interests. No external funding.

Code: <https://github.com/bsharma173860-oss/d3-research>

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Appendix: Verification Code

```
import math
G=6.67430e-11; c=2.99792458e8
hbar=1.054571817e-34; k_B=1.380649e-23; l_P=1.616255e-35
# Verify for any Lambda
for Lambda in [1.089e-52, 1e-40, 1e-50, 1e-60]:
    R_dS = c/math.sqrt(Lambda/3)
    T_dS = hbar*c*math.sqrt(Lambda/3)/(2*math.pi*k_B)
    D3 = 2*G*k_B*T_dS*math.log(2)/c**4
    S = math.pi*c**3*R_dS**2/(G*hbar*math.log(2))
    t_H = R_dS/c
    ratio = D3*S/(c**2*t_H)
    print(f"Lambda={Lambda:.0e}: ratio = {ratio:.12f}")
# Output: ratio = 1.000000000000 for all Lambda
```